

REV 2026

Semester I

Lab Course

1.5 Credits

45 Hours

Engineering Drawing with CAD

Course Code 1004 — Complete Polytechnic Revision 2026 Handbook

PROGRAMME(S)

AI, AI & ML, Biomedical Engg, Cloud & Big Data, Cyber Forensics, CS & Tech, CS & Engg, Computer Engg, CS & Engg (AI & ML), Electronics & Comm, Electronics Engg, Electronics Engg (Embedded Systems), Electronics & Computer Engg, IC Design & Fabrication, Instrumentation Engg, IT, Micro Electronics, Automation & Robotics, RPA, Textile Tech

SUBJECT CODE

1004

SEMESTER

I

COURSE TYPE

Lab

TEACHING SCHEME (L:T:P:R)

0:0:3:0

CONTACT HOURS

45

CREDITS

1.5

CIE MARKS

60

ESE MARKS

40

Official Source Declaration

 **Source:** Kerala Polytechnic Revision 2026 Syllabus — Course 1004 (Engineering Drawing with CAD). This handbook is prepared strictly from the official syllabus document provided. All module headings, topics, subtopics, practical outcomes, assessment schemes, and learning resources are derived directly from the source.

 **Verified Inconsistencies:** The syllabus lists the Teaching Scheme as 0:0:3:0 (L:T:P:R) with 45 contact hours and 1.5 credits. The assessment table shows CIE 60 and ESE 40. The self-learning description in the detailed syllabus table (Page 2) is partially cut off in the PDF. The open-ended experiment description states that drawing sheets from Modules I and II may be considered open-ended experiments. The series test schedule is noted as "Series test - I (module II/II)" and "Series test- II (module II/II)" — the module numbering appears to use Roman numerals inconsistently (II vs II). This handbook treats "module II" as Module II and "module III/IV" as Modules III/IV respectively, following the context. The CO-PO matrix is presented with correlation levels; this handbook reproduces it as given. All other content is faithfully extracted.

Course Dashboard

45

CONTACT HOURS

1.5

CREDITS

4

MODULES

60+40

CIE + ESE

Course Introduction

Engineering graphics is the fundamental language of engineers used to visually communicate ideas, designs, and technical information without any language barriers. Its objectives include developing the ability to interpret and create accurate drawings. Understanding the shape, size, and relationship of components, and applying standards for clear communication. It also enhances spatial visualization and supports the effective planning and manufacture of engineering products. Introduction of CAD software and practicing the development of

drawings along with Engineering graphics give an insight to students regarding development of drawings with accuracy and efficiency without paper usage in engineering design.

Pre-requisites

- Basic Geometry (Secondary Education)

Teaching & Assessment Scheme

L	T	P	S	Self-learning (SS)/Week	Total Hrs/Sem	Credits C	CIE	ESE	Total
0	0	3	0	1.5	45	1.5	60	40	100

L: Lecture (1 credit), T: Tutorial (1 credit), P: Practical (0.5 credit), S: Project (0.5 credit), SS: Self-Learning, CIE: Continuous Internal Evaluation, ESE: End Semester Examination

How to Use This Handbook

1

Understand

Read each module's concepts, definitions, and principles.

2

Visualise

Study the diagrams, SVGs, and worked examples to see the geometry in action.

3

Apply

Practice with the module-wise questions, calculators, and CAD exercises.

4

Revise

Use the quick-revision cards, formula bank, and model paper before exams.

 **Tip:** Use the Print / Save as PDF button to save a printable PDF of the entire handbook. The Search box at the bottom helps you quickly find topics across all modules.

Course Outcomes & CO-PO Mapping

Course Objectives

1. To familiarize the language of graphics which is used to express ideas, convey instructions while carrying out engineering jobs.
2. Provide skills to translate ideas into sketches and to draw and read various engineering curves, conic sections and the generation of 3D solid surfaces into plane areas.
3. To apply basic skill to develop projection of points, lines, orthographic views and sectional views of objects.
4. To familiarize the use of CAD software.
5. To acquire skill in developing orthographic and sectional views using CAD.

Course Outcomes (CO)

Course Outcomes and Bloom's Taxonomy Level

CO	Course Outcome	Bloom's Level
CO1	Construct geometrical figures and illustrate development of surfaces.	Apply
CO2	Interpret projections of points and lines, orthographic projections and sectional views.	Apply
CO3	Familiarization in using CAD software.	Apply
CO4	Develop orthographic projections and sectional views in CAD.	Apply

CO-PO Mapping

Course Articulation Matrix

Course Outcomes	PROGRAM OUTCOMES										
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	3	-	-	-	-	-	-	-	-
CO2	3	3	2	-	-	-	-	-	-	-	-
CO3	3	-	2	3	-	-	-	-	-	-	-
CO4	3	2	3	3	-	-	-	-	-	-	-

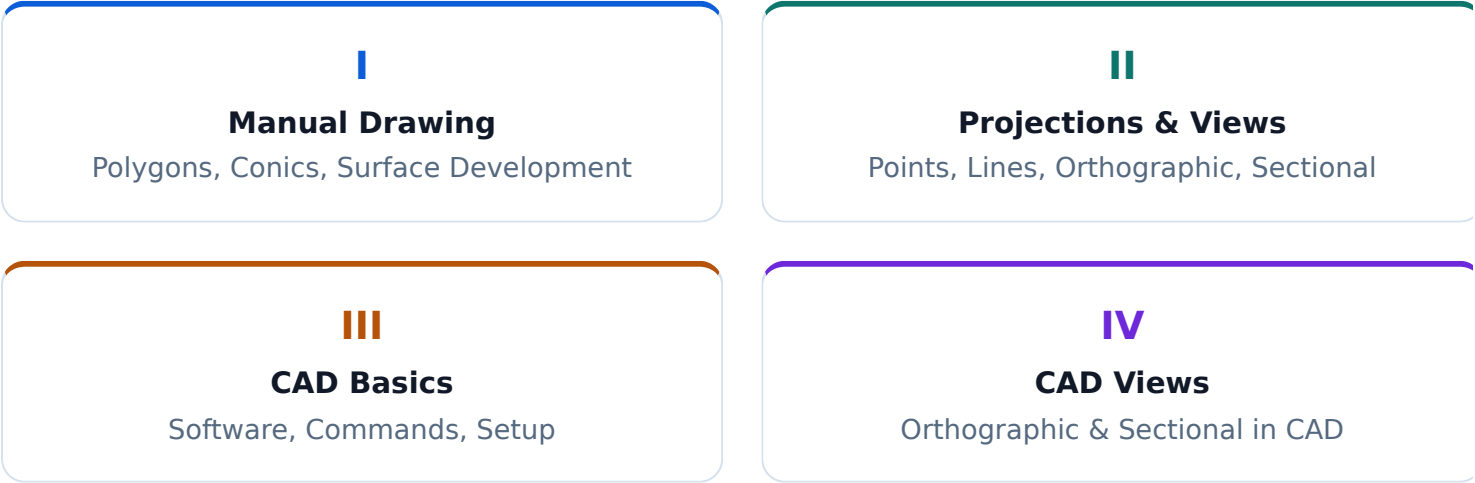
Legends: 1 - Low; 2 - Medium; 3 - High; No Correlation - “-”

Curriculum Coverage Map

Module-wise Topics, Hours, CO & Handbook Anchor

Module	Topics	Hours	CO	Anchor
I	Regular Polygons, Conic Sections, Development of Surfaces	9	CO1	Module I
II	Basic Projections, Projections of Points & Lines, Orthographic & Sectional Views	15	CO2	Module II
III	CAD Software Familiarization & Basic Commands	9	CO3	Module III
IV	Orthographic & Sectional Views in CAD	9	CO4	Module IV

How the Modules Connect



MODULE I

Drawing — Regular Polygons, Conic Sections & Development of Surfaces

Learn to construct regular polygons, draw conic sections (ellipse, parabola, hyperbola), and develop surfaces of 3D solids into 2D patterns.

Hours: 9

CO: CO1

Bloom: Apply

Type: Manual Drawing

1.1 Regular Polygons — Construction of Pentagon & Hexagon▼

Definition: A regular polygon is a closed plane figure with all sides equal in length and all interior angles equal. Common regular polygons used in engineering drawing include pentagon (5 sides), hexagon (6 sides), heptagon (7 sides), and octagon (8 sides).

Key Terminology:

- **Side (s):** Length of one side.
- **Interior angle:** $(n - 2) \times 180^\circ / n$ where n is the number of sides.
- **Exterior angle:** $360^\circ / n$
- **Circumradius (R):** Radius of the circle passing through all vertices.
- **Apothem (r):** Distance from centre to midpoint of a side.

For a regular polygon with n sides and side length s:
Circumradius $R = s / (2 \times \sin(180^\circ/n))$ | Apothem $r = s / (2 \times \tan(180^\circ/n))$

Construction of a Regular Pentagon (Side Length Given)

Method: General method using circumcircle

1. Calculate the circumradius using $R = s / (2 \times \sin(36^\circ))$.
2. Draw a circle with radius R and mark the centre O.
3. Draw a vertical diameter AB.
4. Divide the circle into 5 equal parts (72° each) using a protractor or compass.
5. Join the consecutive points to form the pentagon.

WORKED EXAMPLE

Given: Side length = 40 mm. Construct a regular pentagon.

$R = 40 / (2 \times \sin 36^\circ) = 40 / (2 \times 0.5878) = 40 / 1.1756 \approx 34.03 \text{ mm.}$

Draw circle of radius 34.03 mm, divide into 5 equal parts (72°), join the points.

Definition: Conic sections are curves obtained by intersecting a right circular cone with a plane. The three main conic sections are:

- **Ellipse:** Plane cuts all generators of the cone (angle between plane and axis $>$ semi-vertical angle).
- **Parabola:** Plane is parallel to one generator (angle = semi-vertical angle).
- **Hyperbola:** Plane cuts both nappes (angle $<$ semi-vertical angle).

Ellipse — Rectangular Method

This method is used when the major and minor axes are given.

1. Draw the major axis AB and minor axis CD intersecting at O.
2. Construct a rectangle with sides equal to AB and CD.
3. Divide the rectangle into equal parts horizontally and vertically.
4. Draw lines from the ends of the axes through the division points.
5. Mark the intersections and draw a smooth curve through them.

Ellipse — Concentric Circle Method

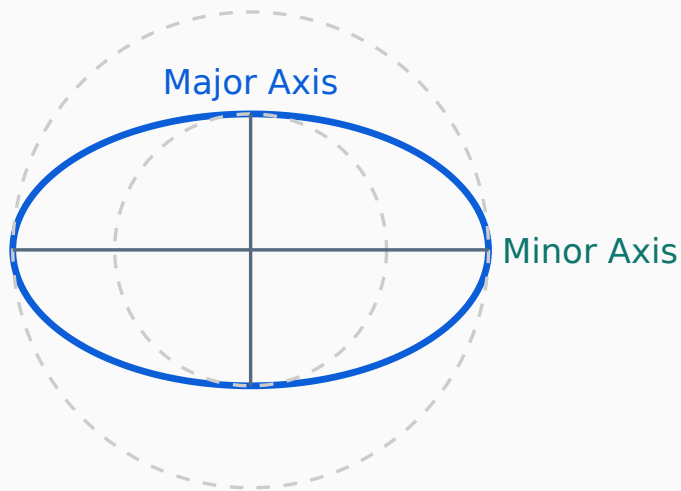
This method uses two concentric circles with diameters equal to the major and minor axes.

1. Draw two concentric circles with diameters equal to the major axis (outer) and minor axis (inner).
2. Divide the circles into 12 equal parts (30° intervals).
3. From each division on the outer circle, draw a vertical line downward.
4. From each division on the inner circle, draw a horizontal line outward.
5. The intersections of these lines give points on the ellipse.
6. Join the points with a smooth curve.

WORKED EXAMPLE — CONCENTRIC CIRCLE METHOD

Given: Major axis = 80 mm, Minor axis = 50 mm.

Draw circles of radii 40 mm and 25 mm. Divide into 12 parts. Project vertically from outer circle and horizontally from inner circle. Intersection points trace the ellipse.



Concentric Circle Method — Ellipse

Ellipse construction using the concentric circle method. The outer circle radius = half major axis, inner circle radius = half minor axis.

Tip: The ellipse is a closed curve with two axes of symmetry. The sum of distances from any point on the ellipse to the two foci is constant and equal to the major axis length.

Malayalam support: കോൺ സെക്ഷൻസ് (Conic Sections) — കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ്. കോൺ സെക്ഷൻസ് (Ellipse), പാരബോള (Parabola), ഹൈപ്പർബോള (Hyperbola) കോൺ സെക്ഷൻസ്. കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ് കോൺ സെക്ഷൻസ്.

1.3 Development of Surfaces — Rectangular Prism & Cylinder▼

Definition: Development of a surface is the process of unfolding a 3D solid's surface into a 2D plane (a net). It is used in sheet metal work, packaging, and fabrication to determine the exact shape and size of material needed.

Key Principle: The developed surface must have the same area as the original surface, and the edges of the development correspond to the edges of the solid.

Development of a Rectangular Prism

A rectangular prism (cuboid) has 6 rectangular faces. Its development consists of 4 side rectangles arranged in a strip, plus 2 end rectangles (top and bottom).

1. Draw the four side faces in a row: each face has height = prism height, width = corresponding side length.
2. Attach the top and bottom faces to the appropriate edges.

3. Add fold lines (dashed) and tabs if required.

WORKED EXAMPLE

Given: Prism dimensions: length = 50 mm, width = 30 mm, height = 40 mm.

Development: Four rectangles of size 50×40, 30×40, 50×40, 30×40 arranged in a strip.
Top and bottom: 50×30 each attached to the top and bottom edges.

Development of a Cylinder

A cylinder consists of a curved lateral surface and two circular bases.

- **Lateral surface:** Develops into a rectangle of height = cylinder height and width = circumference = $2\pi r$.
- **Bases:** Two circles of radius r attached to the top and bottom edges of the rectangle.

$$\text{Lateral surface area} = 2\pi rh \quad | \quad \text{Total surface area} = 2\pi r(r + h)$$

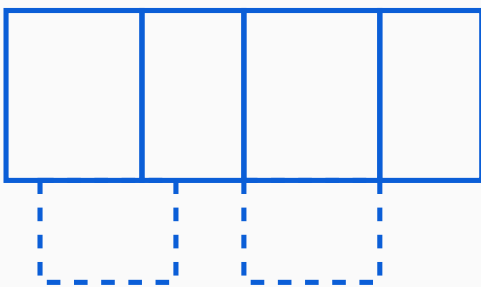
WORKED EXAMPLE

Given: Cylinder radius $r = 20$ mm, height $h = 60$ mm.

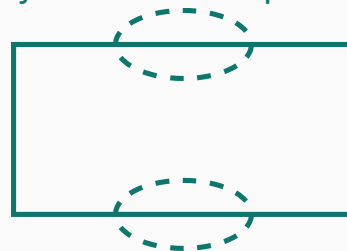
Lateral development: rectangle of width (circumference) = $2\pi \times 20 = 125.66$ mm, height = 60 mm.

Bases: two circles of radius 20 mm.

Prism Development



Cylinder Development



$$\text{Rectangle} = 2\pi r \times h \quad | \quad \text{Circles} = \pi r^2$$

Development (net) of a rectangular prism (left) showing side faces and top/bottom faces, and a cylinder (right) showing lateral rectangle and circular bases.

Malayalam support: പരമാവധി വിസ്തൃതി (Development of Surfaces) — 3D വസ്തുവിന്റെ വിസ്തൃതി 2D രൂപ വിസ്തൃതികളുടെ തുകയാണ്. പരമാവധി വിസ്തൃതി കണ്ടുപിടിക്കാൻ പരമാവധി വിസ്തൃതി (prism) വസ്തുവിന്റെ 6 വിസ്തൃതികൾ, പരമാവധി വിസ്തൃതി കണ്ടുപിടിക്കാൻ പരമാവധി വിസ്തൃതി 2 വിസ്തൃതികൾ വിസ്തൃതികളാണ്.

Drawing – Basic Projections, Points, Lines, Orthographic & Sectional Views

Understand first-angle and third-angle projections, draw projections of points and lines, and develop orthographic and sectional views of engineering objects.

Hours: 15

CO: CO₂

Bloom: Apply

Type: Manual Drawing

2.1 Theory of Projection — First Angle & Third Angle▼

Definition: Projection is the method of representing a 3D object on a 2D plane by projecting its points along parallel lines (orthographic projection) or from a point (perspective projection). Engineering drawing uses **orthographic projection**.

Two main systems:

- **First Angle Projection (European system):** Object is placed in the first quadrant (between the observer and the planes). The view is projected onto the plane behind the object.
- **Third Angle Projection (American system):** Object is placed in the third quadrant (between the planes and the observer). The view is projected onto the plane in front of the object.

Comparison of First Angle and Third Angle Projection

Feature	First Angle	Third Angle
Object position	Between observer and plane	Between plane and observer
View placement	Front view above, top view below	Front view below, top view above

Feature	First Angle	Third Angle
Symbol	Concentric circles (European)	Truncated cone (American)
Commonly used	Europe, Asia (including India)	USA, Canada

Projection Symbols



First Angle



Third Angle

Standard projection symbols: First Angle (concentric circles) and Third Angle (truncated cone).

 **Tip:** In India, the First Angle Projection method is standard for engineering drawings. Always check the projection symbol on the drawing sheet.

Malayalam support: പ്രോജക്ഷൻ (Projection) — 3D വസ്തുവിനെ 2D പ്രതലത്തിൽ പ്രതിബിംബിപ്പിക്കുന്ന പ്രക്രിയ. പ്രോജക്ഷൻ രീതി (പ്രോജക്ഷൻ സിസ്റ്റം), പ്രോജക്ഷൻ കോഡ് (പ്രോജക്ഷൻ കോഡ്) എന്നിവ ഉപയോഗിച്ച് പ്രോജക്ഷൻ സിസ്റ്റം പ്രതിബിംബിപ്പിക്കുന്നു.

2.2 Projections of Points & Lines▼

Projection of Points: A point is a zero-dimensional element. Its projection is a point on the reference plane. The position of the point relative to the Horizontal Plane (HP) and Vertical Plane (VP) determines its projection.

- **First quadrant:** Point is above HP and in front of VP.
- **Second quadrant:** Point is above HP and behind VP.
- **Third quadrant:** Point is below HP and behind VP.
- **Fourth quadrant:** Point is below HP and in front of VP.

Projection of Lines: A line is defined by two points. Its projections are obtained by projecting the two endpoints.

- **Parallel to both planes:** True length appears in both views.
- **Parallel to one plane, inclined to the other:** One view shows true length, the other shows foreshortened length.
- **Inclined to both planes:** Both views show foreshortened lengths.

Projection of Lines — Key Cases (First Quadrant)

Line Position	Front View	Top View
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Line Position	Front View	Top View
Parallel to HP & VP	True length (horizontal)	True length (horizontal)
Parallel to HP, inclined to VP	Foreshortened	True length
Parallel to VP, inclined to HP	True length	Foreshortened
Inclined to both HP & VP	Foreshortened	Foreshortened

WORKED EXAMPLE — PROJECTION OF A LINE

Given: Line AB, length 60 mm, parallel to HP and inclined 30° to VP. End A is 20 mm above HP and 15 mm in front of VP.

Front View: Project A' at 20 mm above XY. Draw a line of length $60 \times \cos 30^\circ = 51.96$ mm at the given angle.

Top View: Project A at 15 mm below XY. Draw the true length 60 mm at 30° to the reference line.

 **Tip:** Always start with the projection of the endpoint that has the most constraints. Use construction lines lightly in pencil. The top view shows the line's inclination to HP; the front view shows inclination to VP.

Malayalam support: ലൈനുകളുടെ പ്രൊജക്ഷൻ — ലൈൻ AB ന്റെ നീളം 60 mm ആണ്, HP-യ്ക്ക് സമാന്തരവും VP-യ്ക്ക് 30° കോണിൽ ചരിഞ്ഞതുമായാണ്. A ക്ക് 20 mm മുകളിലും 15 mm മുന്നിലും സ്ഥാനം ഉണ്ട്. **Front View:** XY രേഖയിൽ 20 mm മുകളിൽ A' പ്രൊജക്ട് ചെയ്യുക. $60 \times \cos 30^\circ = 51.96$ mm നീളം വരയ്ക്കുക. **Top View:** XY രേഖയിൽ 15 mm താഴെ A പ്രൊജക്ട് ചെയ്യുക. 60 mm നീളം 30° കോണിൽ വരയ്ക്കുക.

2.3 Orthographic Projections & Sectional Views of Objects▼

Orthographic Projection: The representation of a 3D object using multiple 2D views — typically front view, top view, and side view. Each view shows the object as seen from a specific direction.

- **Front View:** Projected on the Vertical Plane (VP).
- **Top View:** Projected on the Horizontal Plane (HP).
- **Side View:** Projected on the Profile Plane (PP).

Sectional Views: When an object has internal features (holes, cavities, ribs), a **sectional view** is used to show the internal details. The object is imagined as cut by a cutting plane, and the portion in front of the cutting plane is removed.

- **Cutting plane:** A phantom line (chain line with thick ends) showing where the cut is made.
- **Section lines (hatching):** Thin equidistant lines drawn at 45° on the cut surface.
- **Types:** Full section, half section, offset section, revolved section.

 **Tip:** In orthographic projection, align the views properly — front view above top view, and side view to the side. Use hidden lines (dashed) for edges that are not visible in that view.

 **Common Mistake:** Forgetting to draw hidden lines in orthographic views, or drawing section lines (hatching) on non-sectioned areas. Also, cutting plane lines should be clearly shown with arrowheads indicating the direction of viewing.

Malayalam support: മെറ്റീരിയലുകളുടെ ഓരോ ഭാഗത്തും — 3D മെറ്റീരിയലുകളെ പ്രതിനിധീകരിക്കുന്നു, മെറ്റീരിയലുകൾ, മെറ്റീരിയലുകൾ 2D മെറ്റീരിയലുകളെ പ്രതിനിധീകരിക്കുന്നു. മെറ്റീരിയൽ വ്യക്തിത്വം (Sectional view) മെറ്റീരിയലുകളെ പ്രതിനിധീകരിക്കുന്നു മെറ്റീരിയലുകളെ പ്രതിനിധീകരിക്കുന്നു.

 **Module II Summary:** First-angle projection is used in India. Points are projected based on their quadrant. Lines show true length when parallel to a plane. Orthographic views (front, top, side) describe 3D objects. Sectional views reveal internal features using a cutting plane and hatching.

MODULE III

Lab — CAD Software Familiarization & Basic Commands

Learn to use CAD software — open/save drawings, draft settings, draw commands, modify commands, dimensioning, and text.

Hours: 9

CO: CO3

Bloom: Apply

Type: CAD Lab

3.1 Introduction to CAD — Interface, Commands & Settings▼

Computer-Aided Drafting (CAD): The use of computer software to create, modify, analyse, and optimise designs. CAD replaces manual drafting with digital tools that improve accuracy, efficiency, and ease of modification.

Advantages of CAD over Manual Drafting

- **Accuracy:** CAD allows precise measurements and alignment.
- **Efficiency:** Repeatable commands, copy-paste, and libraries save time.
- **Modification:** Edit drawings easily without redrawing.
- **Storage:** Digital files are easy to store, share, and version-control.
- **Standardisation:** Layers, linetypes, and text styles ensure consistency.

Basic CAD Commands

Common CAD Commands

Category	Command	Purpose
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Category	Command	Purpose
Draw	LINE	Draw straight line segments
	CIRCLE	Draw a circle (centre + radius / diameter)
	RECTANGLE	Draw a rectangle by specifying corners
	POLYGON	Draw a regular polygon (specify sides and centre)
Modify	ERASE	Delete selected objects
	COPY	Duplicate objects
	MIRROR	Create a mirrored copy
	OFFSET	Create parallel copies at a specified distance
	TRIM	Cut objects at intersections
Settings	GRID / SNAP	Snap to grid points for precise placement
	OSNAP	Object snap — snap to endpoints, midpoints, centres, etc.
	DIMENSION	Add dimensions (linear, aligned, radius, diameter)

CAD Workflow

- 1. **Setup:** Set units, limits, grid, and snap settings.
- 2. **Draw:** Use draw commands to create the geometry.
- 3. **Modify:** Edit using modify commands.
- 4. **Annotate:** Add dimensions, text, and notes.
- 5. **Save:** Save the drawing as a .dwg file or export as PDF.

 **Tip:** Use keyboard shortcuts for common commands (e.g., L for LINE, C for CIRCLE, E for ERASE, TR for TRIM). This speeds up your workflow significantly.

Malayalam support: CAD (Computer-Aided Drafting) — കമ്പ്യൂട്ടർ സഹിതം ഡ്രാഫ്റ്റിംഗ്. LINE, CIRCLE, RECTANGLE വരയ്ക്കാനും, ERASE, COPY, MIRROR, TRIM മാറ്റിക്കൊണ്ടും ഉപയോഗിക്കുന്നു. GRID, SNAP, OSNAP എന്നിവ കൃത്യതയ്ക്കായി സഹായിക്കുന്നു. DIMENSION and text add annotations.

 **Module III Summary:** CAD software replaces manual drafting with digital tools. Key commands include LINE, CIRCLE, RECTANGLE, POLYGON (draw), and ERASE, COPY, MIRROR, OFFSET, TRIM (modify). Settings like GRID, SNAP, OSNAP help with precision. Dimensioning and text add annotations.

MODULE IV

Lab — Orthographic & Sectional Views in CAD

Develop orthographic projections and sectional views of engineering objects using CAD software.

4.1 Creating Orthographic & Sectional Views in CAD

This module applies the manual drawing skills from Module II using CAD software. The same principles of orthographic and sectional views are used, but with digital tools.

Step-by-Step Procedure in CAD

- 1. **Analyse the object:** Identify the principal views (front, top, side) and the cutting plane for sectional views.
- 2. **Set up layers:** Create layers for construction lines, object lines, hidden lines, centre lines, section lines, and dimensions.
- 3. **Draw the front view:** Use LINE, CIRCLE, RECTANGLE, and other draw commands to create the front view geometry.
- 4. **Project to top view:** Use projection lines (using OFFSET, COPY, or construction lines) to create the top view aligned below the front view.
- 5. **Project to side view:** Similarly, project the side view from the front view.
- 6. **Add hidden lines:** Use dashed linetypes for edges not visible in each view.
- 7. **For sectional views:** Draw the cutting plane line, then hatch the cut area using the HATCH command with 45° lines.
- 8. **Dimension and annotate:** Add dimensions and text notes.
- 9. **Plot/Print:** Set up the layout, scale, and plot to PDF or print.

CAD Layers for Engineering Drawings

Layer Name	Colour	Linetype	Lineweight	Purpose
Construction	Gray	Continuous	0.25 mm	Construction/projection lines
Object	Black/White	Continuous	0.50 mm	Visible edges
Hidden	Red	Dashed	0.25 mm	Hidden edges
Centre	Blue	Centre (long dash-short dash)	0.25 mm	Centre lines
Section	Green	Continuous	0.25 mm	Section/hatch lines
Dimension	Cyan	Continuous	0.25 mm	Dimensions and text

Important: Always use layers for different line types. This allows you to control visibility, plot styles, and edit efficiently. Do not draw everything on a single layer.

Tip: Use the HATCH command with the "ANSI31" pattern (45° lines) for section hatching. Adjust the scale to get the right spacing. For CAD orthographic views, use PROJECT or construction lines to maintain alignment between views.

Malayalam support: CAD-ഓഫ് ലെയറുകൾ ഉപയോഗിക്കുക, ലെയറുകൾ വ്യത്യസ്തമായി — വ്യത്യസ്ത വ്യക്തിത്വങ്ങൾക്ക് വ്യത്യസ്തമായി വരയ്ക്കുക. വ്യത്യസ്ത വ്യക്തിത്വങ്ങൾ വ്യത്യസ്തമായി വരയ്ക്കുക. HATCH കമാന്റ് ഉപയോഗിച്ച് 45° ലൈനുകൾ ഉപയോഗിച്ച് സെക്ഷൻ ഹാച്ചിംഗ്.

 **Module IV Summary:** Use CAD software to create orthographic views (front, top, side) and sectional views. Organise drawing elements using layers (Object, Hidden, Centre, Section, Dimension). Use HATCH for section lines. Project views using construction lines to maintain alignment.

Formula Bank

Regular Polygon — Interior Angle

$$(n - 2) \times 180^\circ / n$$

n = number of sides

Regular Polygon — Exterior Angle

$$360^\circ / n$$

n = number of sides

Circumradius (Polygon)

$$R = s / (2 \times \sin(180^\circ/n))$$

s = side length, n = number of sides

Apothem (Polygon)

$$r = s / (2 \times \tan(180^\circ/n))$$

s = side length, n = number of sides

Hexagon — Circumradius

$$R = s$$

Side length equals circumradius

Cylinder — Lateral Surface Area

$$2\pi rh$$

r = radius, h = height

Cylinder — Total Surface Area

$$2\pi r(r + h)$$

r = radius, h = height

Cylinder — Circumference

$$2\pi r$$

Used for cylinder development

Diagram Library for Rapid Revision

Regular Polygons

Module I — Pentagon and Hexagon construction using circumcircle method.

Conic Sections

Module I — Ellipse using concentric circle method.

Surface Development

Module I — Nets for rectangular prism and cylinder.

Projection Symbols

Module II — First-angle and third-angle projection symbols.

Orthographic Views

Module II — Front, top, and side view alignment.

Sectional Views

Module II — Cutting plane and hatching for sectional views.

Self-Learning Activities (Official)

Module I — Regular Polygon Terminology

1.5 hours

Definition and terminology of regular polygons. Construction of pentagon, hexagon, heptagon, octagon with given side length.

Module I — Conic Sections

1.5 hours

Methodology and terminology of conic sections. Construction of ellipse, parabola, and hyperbola.

Module I — Development of Surfaces

1.5 hours

Procedure familiarisation for engineering applications of surface development.

Module II — Theory of Projection

1.5 hours

Outline the theory of projection, first-angle and third-angle projections. Projection of points and lines.

Module II — Orthographic & Isometric

3 hours

Practice more exercises on orthographic and isometric projection.

Module II — Sectional Views

3 hours

Draw sectional views of various simple engineering objects. Practice more exercises.

Module III — CAD Basics

1.5 hours

Introduction to Computer Aided Drafting: applications, advantages, hardware/software requirements, CAD basics.

Module III — Draw & Modify Commands

1.5 hours

Limits of drawing, units, grid, snap, osnap, co-ordinates, ortho mode. Draw and modify commands.

Module III — Properties & Dimensioning

1.5 hours

Working with CAD: properties of lines (colour, line weight, line type), dimensions and text on drawings.

Module IV — Orthographic Views in CAD

1.5 hours

Developing orthographic views of simple components with dimensions and text. More practice exercises.

Module IV — Sectional Views in CAD

1.5 hours

Draw sectional views of simple engineering objects. More practice exercises.

Series Exam Preparation

1.5 hours

Series exam preparation (Modules I & II / Modules III & IV).

Total Self-Learning Hours: 22.5

Assessment Methodology

Assessment Breakdown — Lab

Mode	Attendance	CE	CA1	CA2	OEE	ESE	Total
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Mode	Attendance	CE	CA1	CA2	OEE	ESE	Total
Continuous evaluation	—	10	15	15	10	—	—
Practical test (first half)							
Practical test (second half)	—	—	—	—	—	—	—
Open-ended experiment	—	—	—	—	—	—	—
Practical (Duration 3h)	—	—	—	—	—	40	40
Converted to	—	5	5	10	—	—	—
Total marks	—	15	15	10	—	40	100
CIE: 60 marks (Attendance 5 + CE 15 + CA1 15 + CA2 15 + OEE 10) ESE: 40 marks Total: 100							

Scheme of Evaluation — Practical (CIE)

- **Preparation & Punctuality:** 10 marks
- **Experimental Setup & Procedure:** 10 marks
- **Observation & Data Recording:** 5 marks
- **Analysis & Result Interpretation:** 10 marks
- **Viva Voce / Concept Understanding:** 10 marks
- **Workmanship, Discipline & Teamwork:** 5 marks

Total: 50 marks

Scheme of Evaluation — ESE (Practical)

- **Write-up / Procedure:** 10 marks
- **Experiment Setup & Execution:** 10 marks
- **Observation & Result / Output:** 8 marks
- **Viva Voce:** 8 marks
- **Record:** 4 marks

Total: 40 marks

Open-Ended Experiment (OEE)

- **Originality & Relevance:** 10 marks
- **Clarity of Objectives & Plan:** 10 marks
- **Correctness of Execution & Recording:** 10 marks
- **Analysis, Interpretation & Presentation:** 10 marks
- **Teamwork & Innovation:** 10 marks

Total: 50 marks

Series Test & ESE Pattern

- **Series Test I:** Modules I & II (1.5 hrs)
- **Series Test II:** Modules III & IV (1.5 hrs in CAD)
- **ESE Split-up:**
 - 10 MCQs/Fill-ups (CO1 & CO2) — 10 marks
 - CAD Drawing (CO3 & CO4) — 18 marks
 - Viva Voce — 8 marks
 - Record — 4 marks

ESE Total: 40 marks

 **Note:** Students need to draw exercises from Module II on A2/A4 size sheets. If using A4, include sheet size, margin, title block, and angle of projection. The Module II sheets may be considered as the open-ended project. End semester examination is conducted in CAD only. Students must submit a certified fair record for ESE with figures and construction procedure (Module II, Part A) and CAD drawings (Modules III & IV, Part B).

Module-wise Questions with Answers

Module I — Regular Polygons, Conic Sections, Surface Development

Q1: What is a regular polygon? How do you construct a regular hexagon given the side length?▼

Answer: A regular polygon is a closed plane figure with all sides equal and all interior angles equal.

Construction of hexagon (side length s):

1. Since $R = s$ for a hexagon, draw a circle of radius s .
2. Mark any point on the circle as the first vertex.
3. Using the same radius s , step off 6 equal divisions around the circle (60° each).
4. Join the consecutive points to form the hexagon.

Formula: Interior angle = $(6-2) \times 180^\circ / 6 = 120^\circ$. Exterior angle = $360^\circ / 6 = 60^\circ$.

Q2: Explain the concentric circle method for constructing an ellipse.▼

Answer: The concentric circle method is used to construct an ellipse when the major and minor axes are given.

1. Draw two concentric circles with diameters equal to the major axis (outer circle) and minor axis (inner circle).
2. Divide the circles into 12 equal parts (30° intervals) using radial lines from the centre.
3. From each intersection on the outer circle, draw a vertical line downward.

4. From each intersection on the inner circle, draw a horizontal line outward.
5. The intersections of these vertical and horizontal lines give points on the ellipse.
6. Join the points with a smooth curve.

Key principle: The ellipse is the locus of points whose sum of distances to two foci is constant.

Q3: What is meant by "development of surfaces"? Draw the development of a cylinder.▼

Answer: Development of a surface is the process of unfolding a 3D solid's surface into a 2D plane (a net) without distortion.

Cylinder development:

- **Lateral surface:** A rectangle with height = cylinder height (h) and width = circumference ($2\pi r$).
- **Bases:** Two circles of radius r attached to the top and bottom edges of the rectangle.

Area formulas: Lateral = $2\pi rh$, Total = $2\pi r(r + h)$.

Application: Used in sheet metal work to determine the material needed to make a cylindrical object.

Module II — Projections, Orthographic & Sectional Views

Q4: Distinguish between first-angle and third-angle projection.▼

Answer:

- **First-angle projection:** Object is placed between the observer and the projection plane. The front view is above the top view. Used in Europe and India. Symbol: concentric circles.
- **Third-angle projection:** Object is placed between the projection plane and the observer. The front view is below the top view. Used in USA and Canada. Symbol: truncated cone.

In India, the **first-angle projection** method is standard.

Q5: How do you project a line that is parallel to HP and inclined to VP?▼

Answer: For a line parallel to HP and inclined to VP:

- **Top view (on HP):** Shows the true length of the line, since the line is parallel to HP.
- **Front view (on VP):** Shows the foreshortened length ($\text{length} \times \cos \theta$), where θ is the angle of inclination to VP.

Procedure: First draw the top view with the true length at the given angle to the reference line (XY). Then project the front view vertically, showing the foreshortened length.

Q6: What is a sectional view? Why is it used in engineering drawing?▼

Answer: A sectional view is a drawing that shows the internal features of an object by imagining it cut by a cutting plane. The portion in front of the cutting plane is removed to reveal the interior.

Uses:

- Reveals hidden internal details (holes, cavities, ribs).
- Clarifies complex assemblies.
- Reduces the need for hidden lines.

Key elements: Cutting plane line (phantom line with arrowheads), section lines (hatching at 45°) on the cut surface.

Module III & IV — CAD Software

Q7: List the advantages of CAD over manual drafting.▼

Answer:

- **Accuracy:** Precise measurements and alignment.
- **Efficiency:** Commands like COPY, ARRAY, MIRROR save time.
- **Modification:** Easy to edit without redrawing.
- **Storage:** Digital files are compact and easy to share.
- **Standardisation:** Layers and templates ensure consistency.
- **Reusability:** Components can be reused from libraries.
- **Visualisation:** 3D modelling and rendering capabilities.

Q8: Explain the use of layers in CAD. Name the common layers used in engineering drawing.▼

Answer: Layers in CAD are like transparent overlays that organise different types of drawing elements. Each layer can have its own colour, linetype, and linewidth.

Common layers:

- **Construction:** Construction/projection lines (gray, continuous, thin).
- **Object:** Visible edges (black/white, continuous, thick).
- **Hidden:** Hidden edges (red, dashed).
- **Centre:** Centre lines (blue, long dash-short dash).
- **Section:** Section/hatch lines (green, continuous).

- **Dimension:** Dimensions and text (cyan, continuous).

Using layers makes editing, plotting, and visibility control easier.

Q9: Describe the procedure to create a sectional view in CAD.▼

Answer:

1. Draw the orthographic view of the object.
2. Draw the cutting plane line (phantom line) with arrowheads showing the direction of viewing.
3. Use the TRIM or EXTEND commands to prepare the cut geometry.
4. Use the HATCH command to apply hatching (45° lines, ANSI31 pattern) to the cut surface.
5. Add dimensions and text to complete the sectional view.
6. Place the sectional view on the drawing layout with proper alignment.

Tip: Use a separate layer for the hatch pattern so it can be turned off if needed.

Q10: What are the key settings to configure before starting a CAD drawing?▼

Answer:

- **Units:** Set to millimeters (mm) for engineering drawings.
- **Limits:** Define the drawing boundary.
- **Grid & Snap:** Enable grid display and snap to grid points for precision.
- **Object Snap (OSNAP):** Enable endpoints, midpoints, centres, intersections, etc.
- **Layers:** Create and assign layers for different line types.
- **Text Style:** Set a standard text height and font.
- **Dimension Style:** Set arrowheads, text offset, and precision.

Proper initial setup saves time and ensures consistency throughout the drawing.

Practice Model Paper

Based on the official pattern. This is a practice paper, not an official SITTTTR model question paper.

Course 1004 — Engineering Drawing with CAD

End Semester Examination (ESE) — Practice Paper

Duration: 3 hours | **Total Marks:** 40

Part A — Theory (CO1 & CO2)

Answer any **5** of the following. Each question carries 2 marks. ($5 \times 2 = 10$ marks)

1. Define a regular polygon. Write the formula for its interior angle.
2. Explain the concentric circle method for constructing an ellipse.
3. What is meant by the development of surfaces? Give one engineering application.
4. Distinguish between first-angle and third-angle projection with sketches.
5. How do you project a line that is parallel to VP and inclined to HP?
6. Why are sectional views used in engineering drawing?

Part B — CAD Drawing (CO3 & CO4)

Answer any **1** of the following. Each question carries 18 marks. ($1 \times 18 = 18$ marks)

1. Using CAD software, draw the orthographic views (front, top, and side) of a given isometric object (Figure provided in the exam). Show all dimensions and hidden lines.
2. Using CAD software, draw the sectional view of a given engineering component, showing the cutting plane and hatching. Include all dimensions.

Part C — Viva Voce & Record

- **Viva Voce:** 8 marks — Questions based on concepts from all modules.
- **Record:** 4 marks — Completion and neatness of practical record, certified by faculty.

Total: $10 + 18 + 8 + 4 = 40$ marks

Quick Revision

Module I — Polygons, Conics, Development

- Regular polygon: all sides & angles equal.
- Interior angle = $(n-2) \times 180^\circ / n$; Exterior = $360^\circ / n$.
- Hexagon: $R = s$ (circumradius = side length).
- Ellipse: concentric circle method uses 2 circles (major/minor axes).
- Development: unfolding 3D $\rightarrow$ 2D net.
- Cylinder development: rectangle ($2\pi r \times h$) + 2 circles.

Module II — Projections & Views

- First-angle: object between observer & plane (India).
- Third-angle: plane between object & observer (USA).
- Line parallel to HP $\rightarrow$ true length in top view.
- Line parallel to VP $\rightarrow$ true length in front view.
- Orthographic views: front, top, side.
- Sectional views: cutting plane + hatching (45°).

Module III — CAD Basics

- CAD: Computer-Aided Drafting — digital drawing.
- Advantages: accuracy, efficiency, easy modification.

- Draw commands: LINE, CIRCLE, RECTANGLE, POLYGON.
- Modify commands: ERASE, COPY, MIRROR, OFFSET, TRIM.
- Settings: GRID, SNAP, OSNAP, UNITS, LIMITS.
- Dimensioning: linear, aligned, radius, diameter.

Module IV — CAD Views

- Use layers: Object, Hidden, Centre, Section, Dimension.
- Project views using construction lines.
- HATCH command for section lines (ANSI31 pattern).
- Apply dimensions and text annotations.
- Plot/print with proper scale and layout.
- Save as .dwg or export as PDF.

Common Mistakes to Avoid

- ❌ Forgetting hidden lines in orthographic views.
- ❌ Using the wrong projection method symbol on the drawing sheet.
- ❌ Incorrect hatching (section lines drawn on non-cut areas).
- ❌ Drawing all CAD elements on a single layer (no organisation).
- ❌ Forgetting to set units and limits before starting a CAD drawing.
- ❌ Misaligning front, top, and side views in orthographic projection.
- ❌ Using the wrong formula for polygon circumradius (especially hexagon).

Exam Checklist

- ✅ Know the projection symbols (First-angle vs Third-angle).
- ✅ Practice constructing pentagon and hexagon with given side length.
- ✅ Be able to draw ellipse using both rectangular and concentric circle methods.
- ✅ Understand surface development for prism and cylinder.
- ✅ Practice orthographic projection of simple objects (front, top, side).
- ✅ Practice sectional views with cutting plane and hatching.
- ✅ Know CAD commands (draw, modify, settings, dimensioning).
- ✅ Use layers properly in CAD drawings.
- ✅ Prepare a neat record with all exercises and construction procedures.

Glossary

Key Terms and Meanings

Term	Meaning
Regular Polygon	A closed plane figure with all sides and angles equal.
Circumradius	Radius of the circle passing through all vertices of a polygon.
Apothem	Distance from the centre to the midpoint of a side of a polygon.
Conic Section	Curve obtained by intersecting a cone with a plane (ellipse, parabola, hyperbola).

Term	Meaning
Ellipse	Conic section where the plane cuts all generators of the cone.
Development (of a surface)	Unfolding a 3D surface into a 2D plane (net).
Orthographic Projection	Representing a 3D object using multiple 2D views from perpendicular directions.
First-Angle Projection	Projection method where the object is between the observer and the plane (used in India).
Third-Angle Projection	Projection method where the plane is between the observer and the object (used in USA).
Sectional View	A view showing internal details by cutting the object with a plane.
Cutting Plane	A phantom line showing where the object is cut for a sectional view.
Hatching	Thin 45° lines drawn on the cut surface in a sectional view.
CAD (Computer-Aided Drafting)	Software used to create and modify digital engineering drawings.
Layer	An organisational tool in CAD that groups drawing elements with common properties.
OSNAP (Object Snap)	CAD setting that snaps to specific points (endpoint, midpoint, centre, etc.).
HATCH	CAD command used to fill an area with a pattern (e.g., section lines).

Learning Resources

Textbooks

Recommended Textbooks

Sl. No.	Title	Author	Publication
1	Engineering Graphics with AUTOCAD	P I Varghese	VIP Publishers
2	Engineering Graphics	K. C John	PHI Learning Private Limited
3	Engineering Graphics unique methods easy solutions	K N Anilkumar	Adhyuth Narayan Publishers
4	Engineering Drawing with CAD Applications	N. D. Bhatt & V. M. Panchal	Charotar Publishing House

References

Reference Books

Sl. No.	Title	Author	Publication
1	Engineering Drawing	N. D. Bhatt	Charotar Publishing House, Anand, Gujarat
2	Engineering Graphics with AutoCAD	D.M. Kulkarni, A.P. Rastogi, A.K. Sarkar	PHI Learning Private Limited
3	Engineering Drawing	M. B. Shah and B.C. Rana	Pearson Publications

Web Resources

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Major Equipment (for a batch of 30 students): 30 Desktop Computers (Intel i3/i5, 8GB RAM, 500GB HDD/SSD, Ubuntu 22.04 / Windows 10).

Major Software: Suitable version of CAD software (open access/licensed) — 30 licenses.